

Living with

Robi



LIVING WITH ROBI: LEARN HOW TO INTERACT WITH YOUR NEW ROBOT FRIEND

Robi was designed as a companion robot, entertaining you every day by singing, dancing and playing games. He understands more than 200 words, and the following pages give you directions on how to get the most out of your very own Robi!

Robi's creator, the visionary Japanese roboticist and professor Tomotaka Takahashi, believes that one day humanoid robots will live alongside us, and will play a big part in our daily lives. These robots will look like us, talk like us, behave like us and interact with us as naturally as we do with each other, and Robi is one of the very earliest of these 'companion robots' for the 21st

century – built entirely by you in your own home!

Robi can be seen as an entertainment robot – just take the time to read this guide and soon he'll be playing games with you, chatting away to you and generally brightening up your life. And while Professor Takahashi also sees future robots doing jobs for us, like keeping our homes secure and doing

the housework, this is not the role for which Robi was designed. Robi will, however, 'play-act' such tasks – for example, he will pretend to be a 'security robot' or offer to 'clean up' for you, but these are intended to be a bit of fun rather than a practical service. So don't worry if you give the wrong password and Robi says he is 'calling security'!

STARTING ROBI FOR THE FIRST TIME

We explained in full what happens when you turn your Robi on for the first time, but as part of this *Living with Robi* guide, we will provide another brief run-through here.

When Robi is first turned on, he will go through a unique 'waking up' process. He will do this only once, so if you have been building your Robi with friends or family, you may want to make sure everyone is with you when you turn him on for the first time. To ensure this waking up process goes smoothly, follow the steps below:

- Before turning Robi's power on, make sure the final micro SD card, supplied in Stage 70, is inserted correctly.
- Make sure the battery is fully charged.
- Make sure Robi is not still connected to his charger or charging seat.
- Lay Robi out gently on a smooth, flat surface. Arrange him in the posture described in Stage 70.
- Turn Robi's power on, via the switch on his scarf.
- Wait patiently for a few moments, then watch as Robi's eyes light up and he starts to move!
- Robi will then begin asking a series of questions, which you must answer with only 'yes' or 'no'.
- Speak clearly and give Robi a few seconds to recognise your answer – do not repeat it over and over as this can confuse his voice-recognition software.
- Once the waking-up process is complete, Robi's eyes will pulsate gently, letting you know he is ready to receive his first prompt!



Remember: Robi only goes through this waking-up process the very first time he is switched on, so make sure anyone you want to experience it is with you!

USING SPEECH RECOGNITION

Robi is equipped with specifically designed speech-recognition technology, allowing him to understand your voice. The effectiveness of this technology depends on a number of factors, such as the distance of the speaker or background noise, so read the points below to get the most out of your interactions with Robi.

SPEAKING TO ROBI

When you talk to Robi, always speak in clear audible tones. If you speak too softly, or shout, the technology will struggle to pick up each word. Speak naturally and with your typical intonation – Robi will struggle to understand if the pitch of the phrase is too high or inflected.



ROBI'S DIFFERENT RESPONSES

Robi will respond to many prompts with a variety of different answers – see if you can count them all!



GIVE ROBI 2-3 SECONDS TO RESPOND

It is completely normal for Robi to 'think' about each phrase he hears for a few seconds, as his speech-recognition board does its work. If he does not reply or react after three seconds, repeat the phrase clearly.

ROBI'S MOVEMENTS

Sometimes, Robi's response to a prompt will be physical, and again, these will vary. Robi can walk, dance, stand up and sit down. To make him as humanlike as possible, he will also shift very slightly even when not acting out a response, rather than stand motionless.

'HMM?'

As well as shifting very slightly when standing 'still', Robi will sometimes murmur something. Again, this is to make him as humanlike as possible, and is not a question.



ROBI'S EYES

The LEDs on Robi's eyes act in different ways, depending on which 'mode' he is in. When in normal mode, when he is waiting for a prompt, they will pulsate yellow/orange slowly; when in a 'special' mode, such as during a quiz or a game, they will pulsate much faster. More on these modes is found later in this guide.

Before turning Robi's power on

Charge the battery, and set him on a smooth, flat surface.

ROBI'S MODES

To enable you to enjoy a range of interactions with your Robi, he is programmed to have normal and special 'modes', which are explained here.

When you turn Robi on each time, he will be automatically set to his **normal mode**. This means he is ready to receive any of the prompts listed on this page. Some of these prompts initiate simple interactions, such as a single question and a single answer; when such an exchange is complete, Robi will revert again to normal mode, waiting for another prompt. When Robi

is in normal mode, his eyes will gently pulsate orange.

Other prompts initiate more sophisticated, layered interactions, such as the games, quizzes and conversations Robi can perform for your entertainment. When one of these is begun, we refer to Robi entering a **special mode**. Now, the next prompt Robi needs to hear to carry on the

game or interaction must come from the listed **special mode prompts** also found in this guide. When Robi is in one of these special modes, his eyes will pulsate more rapidly than usual, in a range of colours.

When in each special mode, Robi will only respond to the selected 'in-game' prompts that relate to that mode. See pages 5-9 for more on this.

NORMAL MODE PROMPTS

SAYING HELLO & GOODBYE, MAKING FRIENDS

Hello Robi
Hey Robi
Hey Mr Robot
Listen Robi
Hiya Robi
Listen up Robi
Nice to meet you*
Good afternoon
Good evening
Good night Robi
Say 'hello'
Goodbye Robi
See you soon
Bye for now
Bye-bye
Introduce yourself
Hello there, who are you?
What's your name?
Tell me about robots
What can you do?
What are you good at?
Are you a boy?
How old are you?
How tall are you?
How much do you weigh?
Please be my friend
We're friends

** In response to the prompt 'Nice to meet you', Robi will extend his right hand for you to 'shake' by way of a greeting. Press down gently on his hand or arm.*

EVERYDAY CONVERSATION

I'll be home early
I'll be home late
I'll be late back
How was your day?
How are you doing?
I'm off out
See you later

Hello I'm home
Hello I'm back

**Possible mode activated:
At home*

It's going to be sunny
It's a sunny day
It's going to rain
It's raining
It's hot today
It's chilly today
The weather's nice today
Let me take your picture
Are you ready?
What is your favourite colour today?
I ate too much
I have to go on a diet
I've gained too much weight
Let's have a drink
Would you like a drink?
I am so happy
That's so cool

That was fun
I had a great day today
I had a bad day today
That went really well
Oh no, I failed
That was amazing
I like it Robi
That is fun
That was fun
I'm in love with someone new
I love you
Do you like me?
What do you think of me?
I have a cold
I had a fight
I'm broke this month
Oh no, Robi
Oh dear, Robi
That's so sad
That's really hard
Cheer me up, will you?
Say something to cheer me up
Can you cheer me up?
I'm sorry Robi
It's OK
It'll be OK
Takahashi Tomotaka
I'm very tired
I'm exhausted
Are you feeling sleepy?
Let's go to bed now

THANKS AND CONGRATULATIONS

You're a good robot
Excellent
Good job Robi
Congratulations
You're smart
You're so cute
You're so cool
That's very good
Thank you Robi

SPECIAL OCCASIONS

I've got an exam
I passed the test
I've got a match
Cheer for me

MOVEMENTS AND ACTIONS

Start walking
Robi, go forwards
Come towards me
Come over here
Robi, come here
Come to me
Come and face me
Robi, go left
Go to the left
Robi, go right
Go to the right
Turn to the left
Turn to the right
Robi, stand up
Time to wake up
Robi, sit down
Robi, sit still
Robi, move back
Do push-ups
Do sit-ups

ENTERTAINMENT AND GAMES

Play some music
Sing something for me
Please dance with me
Dance with me
Shall we dance together?
Do a mime for me
Transform
Let's play the balancing game
Let's play Kaboom**
Say something funny

Do you want to play?
Let's play together
Could you do something for me?

*Possible modes activated:
Various (see p5-9)

Ask me a question
Quiz me Robi

*Mode activated:
Quiz

Let's play soccer
Play soccer with me

*Mode activated:
Soccer

Wave the flags
Let's play the flag game

*Mode activated:
Flag game

Rock-paper-scissors
Let's play rock-paper-scissors

*Mode activated:
Rock-paper-scissors

MOVEMENTS AND ACTIONS

Set an alarm

*Mode activated:
Timer

Please clean up
Please clean up for me

*Mode activated:
Cleaning

Message mode
Please send a message

*Mode activated:
Message mode

Robi, stand guard
Security mode
Home security
Watch the house

*Mode activated:
Home security

TELEVISION COMMANDS

Turn on the TV
Let's watch TV
Turn off the TV
Change the channel
Turn up the volume
Turn down the volume
Turn the volume down

USEFUL PROMPTS

Tell me what you understand (Robi will list some prompts you can say to him)
Tell me some words you know (Robi will list some prompts you can say to him)
Can we stop? (exits selected special modes, see pages 5-9)
Switch off

ENTERING 'SPECIAL MODES'

The prompts in black are the most basic phrases Robi understands – prompts which elicit a single response, after which Robi will

return to normal mode, ready for the next prompt. The boxed-out phrases, however, send Robi into a 'special mode'. These

modes include games or other functions that involve more complicated interactions. These are explained on page 5.

Robi's battery and special modes

The command 'Switch off' will prompt Robi to turn off – but you must also manually push the power switch on his scarf to the OFF position once this is done. When in normal mode, Robi will warn you if his battery is running out of power by saying 'Please charge me...'. However, if he is in a special mode, this will not happen, so there is a chance his power will cut out without warning, depending on the amount of charge currently held. If this happens while he

is walking, balancing or moving, he may fall over, so be aware of how much charge is left.

Other ways to interact with Robi

Not all of Robi's interactions are verbal. The sensors in his eyes pick up on movement – he might choose to turn to look at you if you give him a wave! As well as this, for some interactions you will need to hold his legs so he can do sit-ups, or 'shake his hand': don't be shy, it's all part of the fun!



Robi will need you to hold his legs for him when he does sit-ups.

SPECIAL MODE PROMPTS

As explained on the previous page, some prompts send Robi into 'special modes', such as his quizzes and games. When in a special mode, Robi's eye LEDs will pulsate more rapidly than usual, showing that he is waiting for one of a set of prompts **specific to the special mode he is in**. It is important to note that for the game to work, you must deliver one of these specified

phrases, which are given in sections in this guide. If you can't remember the phrases specific to the game, you can ask 'Tell me what you understand' or 'Tell me some words you know'. Special modes can be exited either by giving the phrases 'Can we stop?', 'Can we stop now' or 'I want to stop now' or by simply turning Robi off at his scarf switch, then turning him back on.



ENTERING SPECIAL MODES

When you give the normal mode prompts 'Let's play together', 'Do you want to play' or 'Can you do something for me?', Robi will decide which game or activity to initiate. This might be a simple interaction, such as Robi doing a short dance, or could mean him entering a special mode, for example if he decides to initiate a quiz.

If you can't remember which special mode prompts you can use, say:

'Tell me some words you understand'
'Tell me some words you know'

SOCCER MODE

Robi is 'right-footed', so will use his right foot to kick the yellow ball.

Activation commands

- Let's play soccer
- Play soccer with me
- Let's play together
- Do you want to play?
- Could you do something for me?

The prompts marked with ● will *always* initiate the soccer mode, while prompts marked ○ are randomised, so only *might* initiate soccer mode – it's up to Robi to decide!

In-game prompts

To make Robi kick the ball (right foot)

Are you ready Robi, shoot! / Get ready, kick!

To 'control' the ball

Robi, go forward
Come towards me
Robi, go left
Go to the left
Robi, go right
Go to the right

Turn to the left
Turn to the right
Robi, move back
Come to me
Come over here

How to position Robi

Robi needs to be standing to begin the soccer game. If he is sitting when the game is initiated, he will stand up.

Game prop

Yellow ball
(supplied Stage 52)

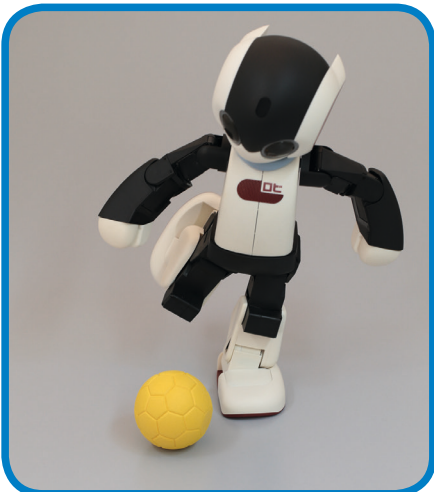


How to play

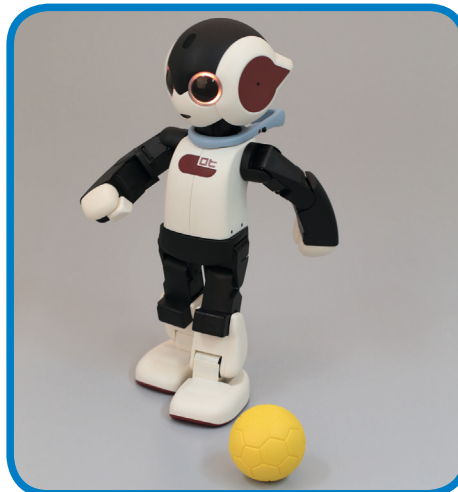
When you begin a game of soccer, Robi will tell you to place the ball at his feet. This should be done in front of his right foot. Then, when you give Robi one of the in-game prompts listed above, he will pull his right leg back then kick the ball forwards. The walking commands for turning left and right, and walking forwards, are then used to 'control' the ball.



The ball must be placed in front of Robi's right foot.



The in-game prompts will tell him to kick the ball.



You can also place the ball away from Robi and get him to walk towards it before kicking.

THE FLAG GAME

Play 'copycat' games with Robi using red and white 'flags'.

Activation commands

- Let's play the flag game
- Wave the flags
- Let's play together
- Do you want to play?
- Could you do something for me?

- will definitely initiate the flag game
- are randomised, so only *might* initiate flag game – again, it's up to Robi!

How to position Robi

Standing or sitting down (Robi will stand up for the game if seated).

Game props

Red and white flags (Stage 69)

In-game prompts

After Robi asks you 'Are you ready?', say:

Go on, start / OK, let's go / OK let's begin / Yeah, Robi / OK, Robi



Remember: the red flag goes in Robi's right hand, and the white flag goes in his left hand.

How to play

To begin, place the red flag in between the fingers and thumb of Robi's right hand, then put the white in his left. When Robi asks if you are ready, you can begin. Then simply follow Robi's instructions!

QUIZ MODE

Robi will ask you more than 10 quiz questions – try and answer them all!

Activation commands

- Give me a quiz
- Quiz me, Robi
- Ask me a question
- Let's play together
- Do you want to play?
- Could you do something for me?

Remember!

- definitely initiate the quiz,
- are randomised, so only *might* initiate the quiz. Robi will decide for you!

In-game prompts

Respond to Robi's questions with:

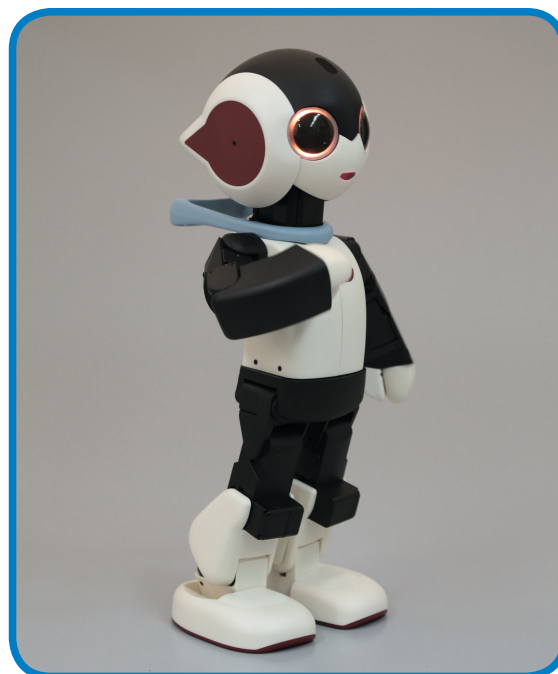
Yeah Robi

No Robi

*Answer Robi only with the above, and note the quiz can not be stopped with a prompt. If you don't feel like a quiz, turn Robi off and on again.

How to position Robi

Standing or sitting down



You can answer all the questions – it doesn't matter if you get one wrong.

How to play

Simply answer Robi's questions with 'Yeah Robi' or 'No Robi'. To continue the quiz, tell Robi again to 'ask me a question', as many times as you like – try to get them all!

KABOOM

Press down on Robi's head or hands – see which makes him go Kaboom!

Activation commands

- Let's play Kaboom
- Let's play together
- Do you want to play?
- Could you do something for me?

Remember!

- definitely initiates Kaboom,
- are randomised, so only *might* initiate Kaboom. Robi will decide for you!

In-game prompts

There are no spoken prompts in this game, just press down on Robi's head or arms.

How to position Robi

Standing or sitting down

How to play

Robi will extend his arms and look up at you: press down on one of his arms or his head – he will either tell you that you are 'safe', or go Kaboom!



Press down gently on Robi's head or hands as shown to play.

ROCK-PAPER-SCISSORS

Play rock-paper-scissors with your Robi.

Activation commands

- Rock-paper-scissors
- Let's play rock-paper-scissors

In-game prompts

After Robi gives his choice, say:
Great! I won
Oh no, I lost
OK, it's a draw

How to position Robi

Standing or sitting down

How to play

When you initiate the game, Robi will start immediately, saying 'Ready? Go – Rock!' (for example). Be ready, and try to time your draw with his. Then it is up to you to tell Robi who won.



Robi's will state his choice verbally, so be ready!

TIMER MODE

Robi can keep time to 1, 3, 5, 7 and 10 minutes for you.

Activation commands

- Set an alarm
- Alarm clock
- Tell me when the time is up
- Set the timer

Programming prompts

After Robi asks 'OK, how long for?', say:
'Set to 1 minute / 3 minutes / 5 minutes
7 minutes / Set to 10 minutes'

How to position Robi

Less than 5 minutes, standing; seated if set for longer.

How to work the timer

Robi will begin counting immediately after you tell him how long to time, and you will see his eyes pulsate green for the duration. He will then tell you when the time is up.

Note: Once he has started counting, Robi cannot be interrupted (though you can manually turn his power off then back on to restore him to normal mode). Also make sure Robi's battery is fully charged before you use the timer mode, and bear in mind that the timer mode is a game function, and is not 100% accurate, so should not be used for very important tasks!

WELCOME HOME MODE

Robi will greet you on your return.

Activation commands

- Hello I'm home
- Hello I'm back

How to position Robi

Standing
Sitting down
On charging chair

How to answer Robi

After Robi welcomes you back, say one of the following to continue the conversation:

Right, time to eat
Come on, dinner time
Let's have dinner
Time for a bath
I'll take a bath

How to use welcome home mode

In this mode Robi will welcome you home, after which you will decide whether to have dinner or take a bath.



HOME SECURITY MODE

Robi will 'watch' the house for intruders, and ask a password when he sees someone.

Activation commands

- Home security
- Robi, stand guard
- Security mode
- Watch the house

Password

When Robi asks for the password, say:

Hello I'm home

* this is the only password Robi will accept, so repeat this phrase only.

How to position Robi

Sitting down. If standing when the command is given, he will sit down.

Using the home security mode

When Robi goes into home security mode, he will sit down and 'watch' the house, using the infrared motion sensors set into his eyes. When he detects movement, he will sound the alarm, call security, or ask for a password. The only way to return Robi to normal mode now is to give the correct password! (Or turn his power off then on to reset him).



TV REMOTE CONTROL MODE

Use Robi to control your TV!

* All the TV commands can be given when Robi is in normal mode, and are subject to compatibility with your TV

Remote control commands

- Turn on the TV
- I want to watch TV
- Let's watch TV
- Turn off the TV
- Change the channel
- Switch channel
- Turn up the volume
- Turn down the volume
- Turn the volume down
- Turn the TV down

How position Robi

Standing
or sitting

How to use the TV remote control mode

Position Robi facing your TV (with his left arm perpendicular to it, so he points at it when sending signals). The infrared transmitter is set into Robi's

forehead, so he must not be sitting higher than the TV, as this will decrease the efficiency of his transmissions. Make sure the mains power to the TV is on before beginning. See Stage 69 for more info on setting and using Robi's TV remote control function.



OK!

TURN ON THE TV

MESSAGE MODE

Using message mode, Robi can deliver a short message for you.

Activation commands

- Message mode
- Please send a message

Messages you can ask Robi to deliver

When Robi asks what the message is, say:

Cheer me up, will you? / Say something to cheer me up / Can you cheer me up? / I had a great day today / I had a bad day today / I passed the test / It didn't go well / It's my birthday / It's Christmas / I've got an exam / I've got a match / I had a fight / It's New Year's Day

The person receiving the message can say:

Any normal mode prompt

How to position Robi

Sitting, or in charging chair (he will sit down if standing)

What is the message?



Deliver messages to your friends and family using message mode.

How message mode works

When message mode is activated, Robi will ask you which message you want to deliver from the ones shown above. Once he has registered your choice, he will go into 'standby mode' – he will then deliver your chosen message to the first person who speaks any normal mode prompt to him.

CLEANING MODE

Robi uses felt pads to clean a surface.

Activation commands

- Please clean up
- Please clean up for me

Prompts to use during cleaning

When Robi has his special felt foot pads fitted and asks if you are ready, say:

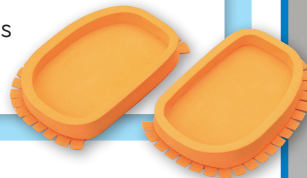
OK Robi, let's go
Go on, start
OK let's begin

How to position Robi

Standing up
(he will stand up if sitting;
if on charging chair he will
need to be rebooted)

Props

Felt foot pads
(supplied in
Stage 34)



How cleaning mode works

When Robi asks you to put his shoes on, simply slip these over the soles of his feet. Then, with Robi in the standing position, give one of the prompts listed above and he will begin 'cleaning' the surface he is on.



Robi cleans by pulling his feet across the surface he is standing on, so this must be a smooth surface for the mode to work.

SING, DANCE AND PLAY WITH ROBI

On this page we list some of the commands you can give to Robi when he is in normal mode, and their responses. There are many fun things you can do with Robi – so try them all! Many of these include lots of physical movements, such as dancing or balancing, so always make sure that Robi's battery is fully charged before you start*. Also be sure to speak clearly when you deliver the prompt**, and to place Robi on a smooth, flat surface in a place where he won't knock anything over or fall!



'SING FOR ME'

When prompted, Robi can sing for you and your friends, playing music from his in-built speaker too!



'LET'S DANCE TOGETHER'

Robi loves to dance, and will play his own music to dance to when you say to him 'Let's dance together!'

'INTRODUCE YOURSELF'

When showing Robi to your friends for the first time, he can introduce himself when prompted!



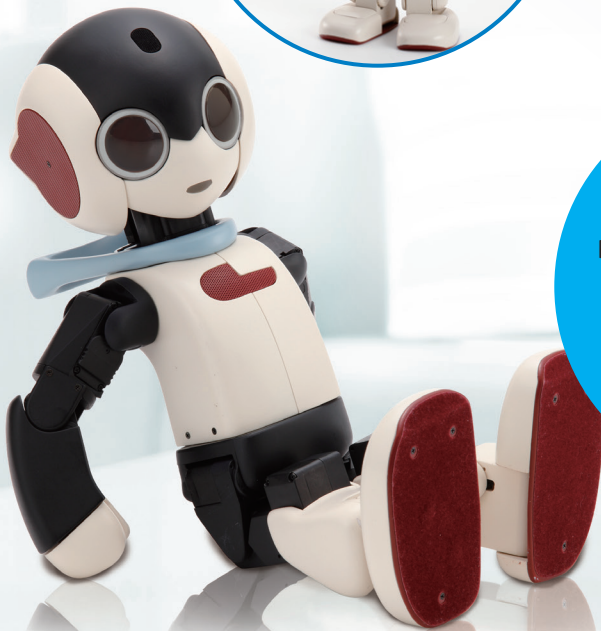
'PLAY THE BALANCING GAME'

Robi can use his servomotors together to perform impressive balancing acts – try to copy the moves he does!



'DO SIT-UPS'

Robi is fighting fit, and able to perform exercises on demand. Ask him to do push-ups or sit-ups and watch him work out!



'TRANSFORM!'

Robi can do impersonations and 'transform' himself into a machine!

* The battery's lifespan will depend on the type of use: for example, physical actions like the balancing game, above, will use more battery than, for example, singing.

** Remember to always speak clearly when you deliver a prompt, using the exact wording given in this guide. Robi's microphone used for receiving prompts is in the front of his head, so position yourself in front of him for the best results. Slowly pulsating yellow/orange eyes means Robi is waiting for a prompt.

ROBI SAFETY AND USAGE TIPS

The *Living with Robi* guide supplied with this pack will give you all the information you need to be able to interact and play with Robi, but this page lists the key points on safety and usage.

Robi start-up and general use

- Robi moves whether he is standing up or sitting down, so always make sure there is enough space around him for him to do so safely. For example, be careful not to place Robi too close to the edge of a tabletop.
- Make sure there are no obstacles around Robi that he may bump into.
- Robi needs a smooth, flat surface to operate on. Surfaces like carpets or tablecloths can cause friction and disturb his movement and balance.
- Robi can stand up from a sitting position on his own, but if on an uneven surface, you may want to hold his scarf for extra support.
- You should never turn the power ON while Robi is charging. If he is on his charging chair and he is switched on, he will tell you to turn him off, remove him from the chair, and turn him on again.
- Do not use Robi while he is charging.

Robi voice recognition

- Robi's voice-recognition technology works best in a quiet environment with little background noise. In ideal conditions he will be able to hear and understand commands from a few metres.
- Robi's microphone for picking up commands is set into the front of his head, so if he has difficulty registering speech, move closer and direct the commands at his forehead.
- Talk to Robi in a clear tone at normal speed – speaking too fast or too softly will affect his ability to register words.
- It may take a few seconds for Robi to reply, depending on the type of command given. Give Robi time to answer, and if some seconds pass without reply, repeat the command.
- A moment of silence is needed before and after each command for the technology to register it successfully.
- Prompts and commands should be delivered exactly as they appear in the *Living with Robi* guide: speak in a clear, level tone with no pauses, and do not alter the wording at all.
- Any deviation from the above will reduce Robi's ability to recognise prompts.

The battery and automatic power-saving mode

- Operation time with a fully charged battery is up to 20 minutes, depending on the types of activities performed.
- If Robi is left standing for some time, he will sit down to conserve power.
- After three minutes of sitting, if no commands are received, Robi will power down to save battery. Turn the power switch to OFF at this point.
- Robi will warn you when his battery is running low, but you can also ask him how much power he has left – see the *Living with Robi* guide for more on this.

- Robi has many 'special functions' – such as quizzes and games – where he will wait for a specific response from you (all of which are explained in full in the *Living with Robi* guide). If he is left too long in this mode, his battery may run out, causing him to topple over. You can turn him off at his switch to avoid this.
- Only use the mains adapter or charging seat provided with the *Build Your Own Robi* series to charge your Robi.
- Do not remove the battery from its plastic coating or tamper with it in any way.
- Keep the battery away from heat sources.
- Never handle the battery with wet hands, and keep it away from any damp areas. (See Stage 38 for more on battery care and use.)

Robi behaviour triggers

- In response to some greetings, Robi will extend his arm for a 'handshake'. When he does this, press down gently on his forearm.
- Robi has a timer mode, but this should not be used for important tasks.
- During some conversations Robi will talk of 'swimming', 'bathing' or firing a laser beam. These are all purely conversational: he must never be put in contact with water, and of course is equipped with no laser technology!
- Similarly, when Robi talks of 'calling security', no actual call will be made.

Other actions/functionality

- Only ever move Robi using his scarf – this is especially important when the power is on.
- It is natural for Robi's CPU, battery and servo motors to warm up a little while in use, but if these heat up too dramatically or rapidly, turn Robi OFF immediately.
- It is possible that a slight hissing sound will be heard as Robi speaks – this is completely normal.
- If Robi falls while walking or balancing, pick him up by the scarf and reset his stance. Also ensure the surface you are using is completely level and smooth.
- More on Robi's balancing is found on page 10.
- Keep Robi out of direct sunlight.
- Do not leave Robi in the standing position when switched off, and if leaving him unused for some time, make sure to store him where there is no stress on his joints, for example in the sitting or lying down position.
- Robi should not be in contact with other electronic objects for long periods of time.
- If Robi's plastic exoskeleton gets dirty, wipe it with a very slightly damp cloth, then rub dry. A mild diluted detergent can be used to dampen the cloth, but on no account should Robi be immersed in water or made too wet.